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Online Algorithms and Sequential Hypothesis Testing for Real-Time Network Anomaly Detection

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Abstract

High-throughput Network Intrusion Detection Systems (NIDS) operating on modern backbone links face severe computational and architectural bottlenecks: packet streams must be inspected in a single pass ($\mathcal{O}(1)$ processing time per packet) and with strictly bounded $\mathcal{O}(1)$ memory per active network flow. Classical two-pass statistical estimators cannot be deployed in this streaming regime because storing unbounded packet histories exhausts router memory, whereas textbook one-pass algebraic reformulations suffer from catastrophic cancellation under finite-precision IEEE 754 floating-point arithmetic. Furthermore, traditional batch hypothesis tests require accumulating a fixed window of observations before evaluating decision rules, introducing unacceptable mitigation latency during fast network attacks.

This paper investigates the mathematical and statistical foundations of real-time network anomaly and port-scan detection by unifying streaming numerical algorithms, asymptotic probability theory, and sequential analysis. First, we perform a rigorous backward and forward error analysis of sample variance computation in finite-precision floating-point arithmetic, deriving the condition number of empirical variance estimation and proving by mathematical induction both Welford's online recurrence and Chan's parallel partition theorem for distributed network probes. Second, we establish the measure-theoretic concentration bounds (Markov's and Chebyshev's inequalities, the First Borel-Cantelli Lemma) underlying the Weak Law of Large Numbers (WLLN), Kolmogorov's Strong Law of Large Numbers (SLLN), and the Lévy-Lindeberg Central Limit Theorem (CLT). Using these asymptotic guarantees alongside Slutsky's Theorem, we construct studentized fixed-window anomaly detectors and derive closed-form expressions for the minimum sample size N_{CLT} required to satisfy prescribed False Positive (α) and False Negative (β) error bounds. Third, to eliminate fixed-window latency, we formulate online port-scan detection as Wald's Sequential Probability Ratio Test (SPRT), known in network security as the Threshold Random Walk (TRW). We derive Wald's stopping boundaries from first principles and apply Gibbs' inequality together with the SLLN to prove that the cumulative Log-Likelihood Ratio exhibits a strictly positive Kullback-Leibler drift under attack, guaranteeing almost sure finite stopping time ($\mathbb{P}(\tau < \infty) = 1$). Empirical benchmarks validate both the numerical stability of Welford's recurrence across eight orders of magnitude of clock offset and the $4\times$ to $6\times$ latency reduction of sequential testing over batch CLT detectors.

1 Introduction and Statistical Traffic Model

Modern enterprise and backbone networks routinely operate at line rates between 10 Gbps and 100 Gbps. At 10 Gbps, a link carrying minimum-sized Ethernet frames (64 bytes plus 20 bytes of framing overhead) processes 14.88 million packets per second, leaving a budget of merely 67.2 nanoseconds per packet. At 100 Gbps, this inter-packet budget shrinks to 6.72 nanoseconds. A Network Intrusion Detection System (NIDS) or edge telemetry probe deployed in such an environment operates under two non-negotiable engineering constraints:

(i) **Single-Pass Execution ($\mathcal{O}(1)$ Time Complexity):** Each arriving packet header can be inspected only once as it traverses the network interface pipeline. Iterative re-scanning of historical packets is computationally infeasible.

(ii) **Bounded Per-Flow State ($\mathcal{O}(1)$ Space Complexity):** Because an edge router simultaneously tracks hundreds of thousands of active source-destination pairs, maintaining raw time series of length n in high-speed SRAM or TCAM memory quickly leads to state exhaustion, especially when an adversary intentionally spoofs or rotates source IP addresses.

Within this streaming computational model, we study a unified operational cybersecurity problem: the real-time statistical detection of reconnaissance *port scanning* (horizontal and vertical TCP probing) and anomalous flow timing behavior. When an external adversary prepares an intrusion, it typically probes the target network by sending TCP `SYN` segments across either a range of destination ports on a single host (vertical scan) or a single service port across an entire subnet (horizontal scan), or by injecting high-rate packet bursts that perturb baseline inter-arrival statistics.

1.1 Probabilistic Formulation of TCP Flows

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a complete probability space modeling the stochastic arrival of network packets at a monitoring interface. For any monitored source IP address, packet arrivals are indexed sequentially by $i \in \mathbb{N}^+ = \{1, 2, \dots\}$. Let $\{\mathcal{F}_n\}_{n\geq 0}$ denote the natural filtration generated by the observed traffic stream, where $\mathcal{F}_0 = \{\emptyset, \Omega\}$ and $\mathcal{F}_n = \sigma\big((X_1, Y_1), \dots, (X_n, Y_n)\big)$ represents all telemetry information available up to the n-th packet arrival. Each packet i generates a bivariate observation (Y_i, X_i):

1. **Continuous Flow Metric ($Y_i \in \mathbb{R}$):** A real-valued telemetry feature such as the packet inter-arrival timestamp t_i, the inter-arrival delay $\Delta t_i = t_i - t_{i-1}$, or the instantaneous packet header rate. Under stationary traffic conditions, $\{Y_i\}_{i=1}^n$ is modeled as a sequence of independent and identically distributed (i.i.d.) random variables with unknown population mean $\mu = \mathbb{E}[Y_i]$ and finite, strictly positive population variance $0 < \sigma^2 = \mathrm{Var}(Y_i) = \mathbb{E}[(Y_i - \mu)^2] < \infty$.

2. **Binary Connection Outcome ($X_i \in \{0, 1\}$):** An indicator random variable associated with the i-th TCP `SYN` connection attempt initiated by the remote

host:

$$X_i = \begin{cases} 0, & \text{if connection } i \text{ succeeds (target returns TCP } \texttt{SYN-ACK}\text{)}, \\ 1, & \text{if connection } i \text{ fails (target returns TCP } \texttt{RST} \text{ or times out)}. \end{cases} \tag{1.1}$$

Conditioned on the true operational mode of the remote host, the sequence $\{X_i\}_{i=1}^n$ consists of i.i.d. Bernoulli(θ) random variables with probability mass function:

$$\mathbb{P}(X_i = x \mid \theta) = \theta^x (1-\theta)^{1-x}, \qquad x \in \{0,1\}, \quad \theta \in (0,1). \tag{1.2}$$

For each individual observation, the expectation and variance are given by $\mathbb{E}[X_i \mid \theta] = \theta$ and $\mathrm{Var}(X_i \mid \theta) = \theta(1-\theta)$.

1.2 Statistical Hypotheses for Network Reconnaissance

The fundamental distinction between benign traffic and reconnaissance scanning lies in the prior knowledge of active network services. Legitimate clients resolve domain names via DNS or follow application-layer links pointing to active servers and open ports. Consequently, under the null hypothesis H_0 (benign host), connection failures occur only sporadically due to transient congestion or server maintenance, resulting in a low baseline failure probability θ_0 (typically $\theta_0 \in [0.05, 0.15]$).

Conversely, an unauthorized scanner lacks an internal map of active hosts and open services. Because the vast majority of the $2^{16} = 65{,}536$ TCP ports and internal subnet addresses are either unallocated or filtered by firewalls, exploratory probes predominantly trigger TCP `RST` responses or silent packet drops. Thus, under the alternative hypothesis H_1 (malicious scanner), the failure probability θ_1 is substantially higher than the baseline ($\theta_1 > \theta_0$, typically $\theta_1 \in [0.50, 0.95]$ for standard scans, and $\theta_1 \in [0.25, 0.45]$ for evasive low-and-slow probes). The NIDS must therefore solve two coupled statistical problems in $\mathcal{O}(1)$ memory:

(a) Maintain numerically accurate running estimates of the sample mean $\bar{y}_n$ and sample variance s_n^2 for continuous flow features $\{y_i\}_{i=1}^n$ without suffering from floating-point overflow or catastrophic cancellation.

(b) Decide between $H_0 : \theta = \theta_0$ and $H_1 : \theta = \theta_1$ with False Positive Rate at most α and False Negative Rate at most β while minimizing the number of observed packets n before blocking a malicious source.

2 Streaming Flow Statistics: Welford's and Chan's Algorithms

Given a stream of real-valued flow measurements $\{y_1, y_2, \ldots, y_n\}$, the unbiased sample mean $\bar{y}_n$ and unbiased sample variance s_n^2 are defined in exact real arithmetic by:

$$\bar{y}_n = \frac{1}{n}\sum_{i=1}^{n} y_i, \qquad s_n^2 = \frac{1}{n-1}\sum_{i=1}^{n}(y_i - \bar{y}_n)^2 = \frac{M_{2,n}}{n-1}, \tag{2.1}$$

where $M_{2,n} = \sum_{i=1}^{n}(y_i - \bar{y}_n)^2$ denotes the centered sum of squares. Computing s_n^2 directly from Eq. (2.1) is a *two-pass* procedure: the first pass over all n observations computes $\bar{y}_n$, and the second pass accumulates the squared deviations $(y_i - \bar{y}_n)^2$. Because storing $\{y_1, \ldots, y_n\}$ requires $\mathcal{O}(n)$ memory, the two-pass definition cannot be executed directly in a streaming NIDS.

2.1 Catastrophic Cancellation in IEEE 754 Floating-Point Arithmetic

To eliminate the second pass, elementary statistics textbooks expand the quadratic term in $M_{2,n}$ to obtain the classical *naive one-pass* algebraic identity:

$$M_{2,n} = \sum_{i=1}^{n}(y_i^2 - 2y_i\bar{y}_n + \bar{y}_n^2) = \sum_{i=1}^{n} y_i^2 - 2\bar{y}_n \sum_{i=1}^{n} y_i + n\bar{y}_n^2 = \sum_{i=1}^{n} y_i^2 - \frac{1}{n}\left(\sum_{i=1}^{n} y_i\right)^2. \quad (2.2)$$

Dividing by $n-1$ yields the one-pass variance estimator:

$$s^2_{n,\text{naive}} = \frac{1}{n-1}\left(S_{2,n} - \frac{1}{n}S_{1,n}^2\right), \qquad \text{where } S_{1,n} = \sum_{i=1}^{n} y_i, \quad S_{2,n} = \sum_{i=1}^{n} y_i^2. \quad (2.3)$$

Although Eq. (2.3) requires only $\mathcal{O}(1)$ memory (storing the triplet $(n, S_{1,n}, S_{2,n})$), its behavior in finite-precision computer arithmetic is disastrous whenever the signal has a large mean offset relative to its standard deviation ($\mu^2 \gg \sigma^2$). In network telemetry, this regime occurs constantly: packet arrival timestamps measured from the UNIX epoch ($t_i \approx 1.77 \times 10^9$ s) with millisecond jitter ($\sigma \approx 10^{-3}$ s) exhibit a ratio $\mu/\sigma \approx 10^{12}$, and high-volume byte counters exhibit similar offsets.

To formalize this failure, recall that an IEEE 754 binary floating-point number is represented as $\text{fl}(x) = \pm(1.b_1b_2\ldots b_{p-1})_2 \times 2^e$ with a precision of p significand bits and unit roundoff (machine epsilon) $\varepsilon_{\text{mach}} = 2^{-p}$ (for IEEE 754 single precision `float32`, $p = 24$ and $\varepsilon_{\text{mach}} \approx 5.96 \times 10^{-8}$; for double precision `float64`, $p = 53$ and $\varepsilon_{\text{mach}} \approx 1.11 \times 10^{-16}$). Under the standard model of floating-point arithmetic, any basic arithmetic operation $\odot \in \{+, -, \times, /\}$ satisfies:

$$\text{fl}(a \odot b) = (a \odot b)(1 + \delta), \qquad |\delta| \le \varepsilon_{\text{mach}}. \quad (2.4)$$

Proposition 2.1 (Condition Number and Error Amplification of Naive Variance)**.** *Let $\{y_i\}_{i=1}^n$ be observations with sample mean $\bar{y}_n = \mu$ and true centered sum of squares $M_{2,n} = (n-1)\sigma^2 > 0$. When $M_{2,n}$ is evaluated in floating-point arithmetic via the uncentered difference $\widehat{M}_{2,n} = \text{fl}\big(\text{fl}(S_{2,n}) - \text{fl}(n\bar{y}_n^2)\big)$, the relative forward error satisfies:*

$$\frac{|\widehat{M}_{2,n} - M_{2,n}|}{M_{2,n}} \le \mathcal{O}(n\,\varepsilon_{\text{mach}})\left(1 + \frac{n}{n-1}\frac{\mu^2}{\sigma^2}\right) \approx \mathcal{O}\left(n\,\varepsilon_{\text{mach}}\frac{\mu^2}{\sigma^2}\right). \quad (2.5)$$

Proof. In exact arithmetic, $S_{2,n} = \sum_{i=1}^{n} y_i^2 = M_{2,n} + n\mu^2 = (n-1)\sigma^2 + n\mu^2$, while the second term equals $B_n = \frac{1}{n}S_{1,n}^2 = n\mu^2$. When accumulated in floating-point arithmetic over n steps, standard backward error analysis of recursive summation

gives $\mathrm{fl}(S_{2,n}) = S_{2,n}(1+\eta_1)$ and $\mathrm{fl}(B_n) = B_n(1+\eta_2)$, where $|\eta_1|, |\eta_2| \le c\,n\,\varepsilon_{\mathrm{mach}}$ for a small constant $c \ge 1$. Subtracting the two quantities yields:

$$\begin{aligned}\widehat{M}_{2,n} &= \big[S_{2,n}(1+\eta_1) - B_n(1+\eta_2)\big](1+\delta_{\mathrm{sub}}) \\ &= (S_{2,n} - B_n) + S_{2,n}\eta_1 - B_n\eta_2 + \mathcal{O}(\varepsilon_{\mathrm{mach}}^2) \\ &= M_{2,n} + \big((n-1)\sigma^2 + n\mu^2\big)\eta_1 - n\mu^2\eta_2 + \mathcal{O}(\varepsilon_{\mathrm{mach}}^2).\end{aligned}$$

Subtracting $M_{2,n} = (n-1)\sigma^2$ and dividing by $M_{2,n}$ gives the relative error:

$$\frac{|\widehat{M}_{2,n} - M_{2,n}|}{M_{2,n}} \le |\eta_1| + \frac{n\mu^2}{(n-1)\sigma^2}\big(|\eta_1| + |\eta_2|\big) \le 2c\,n\,\varepsilon_{\mathrm{mach}}\left(1 + \frac{n}{n-1}\frac{\mu^2}{\sigma^2}\right).$$

When $\mu^2/\sigma^2 \gg 1$, the leading digits of $S_{2,n}$ and B_n are identical and cancel during subtraction (*catastrophic cancellation*), leaving only accumulated rounding noise. In particular, whenever $\frac{\mu}{\sigma} \ge \varepsilon_{\mathrm{mach}}^{-1/2}$, the relative error exceeds 100%, and when $\eta_2 > \eta_1$, the computed variance $\hat{s}^2_{n,\mathrm{naive}}$ becomes strictly negative. □

2.2 Inductive Derivation of Welford's Online Recurrence

To eliminate the uncentered quadratic factor μ^2/σ^2, B. P. Welford (1962) derived a recurrence that updates the centered sum of squares $M_{2,n}$ directly at each arrival y_n by multiplying the deviation from the *old* mean $\bar{y}_{n-1}$ by the deviation from the *new* mean $\bar{y}_n$.

Theorem 2.2 (Welford's Online Recurrence)**.** *Let $\{y_i\}_{i=1}^n$ be any sequence of real numbers. Initialize $\bar{y}_1 = y_1$ and $M_{2,1} = 0$. For every integer $n \ge 2$, the sample mean $\bar{y}_n = \frac{1}{n}\sum_{i=1}^n y_i$ and the centered sum of squared deviations $M_{2,n} = \sum_{i=1}^n (y_i - \bar{y}_n)^2$ satisfy the exact recurrences:*

$$\bar{y}_n = \bar{y}_{n-1} + \frac{y_n - \bar{y}_{n-1}}{n}, \tag{2.6}$$

$$M_{2,n} = M_{2,n-1} + (y_n - \bar{y}_{n-1})(y_n - \bar{y}_n), \tag{2.7}$$

and the unbiased sample variance for $n \ge 2$ is given by $s_n^2 = \frac{M_{2,n}}{n-1}$.

Proof. We proceed by mathematical induction on $n \ge 1$.

Base Case ($n = 1$): By definition, $\bar{y}_1 = \frac{1}{1}y_1 = y_1$ and $M_{2,1} = (y_1 - \bar{y}_1)^2 = 0$, matching the initialization.

Inductive Step ($n-1 \implies n$): Suppose that for $n - 1 \ge 1$, the quantities $\bar{y}_{n-1}$ and $M_{2,n-1} = \sum_{i=1}^{n-1}(y_i - \bar{y}_{n-1})^2$ have been computed exactly. First, we establish the mean recurrence (2.6) by separating the n-th observation from the sum:

$$\bar{y}_n = \frac{1}{n}\sum_{i=1}^n y_i = \frac{1}{n}\left(\sum_{i=1}^{n-1} y_i + y_n\right) = \frac{(n-1)\bar{y}_{n-1} + y_n}{n} = \bar{y}_{n-1} + \frac{y_n - \bar{y}_{n-1}}{n}. \tag{2.8}$$

From Eq. (2.8), we immediately deduce two key algebraic identities linking the pre-update residual $\delta_1 = y_n - \bar{y}_{n-1}$ and the post-update residual $\delta_2 = y_n - \bar{y}_n$:

$$\bar{y}_n - \bar{y}_{n-1} = \frac{y_n - \bar{y}_{n-1}}{n}, \tag{2.9}$$

$$y_n - \bar{y}_n = (y_n - \bar{y}_{n-1}) - (\bar{y}_n - \bar{y}_{n-1}) = \left(1 - \frac{1}{n}\right)(y_n - \bar{y}_{n-1}) = \frac{n-1}{n}(y_n - \bar{y}_{n-1}). \tag{2.10}$$

Combining Eq. (2.9) and Eq. (2.10) yields the fundamental bridge relation:

$$(n-1)(\bar{y}_n - \bar{y}_{n-1}) = \frac{n-1}{n}(y_n - \bar{y}_{n-1}) = y_n - \bar{y}_n. \tag{2.11}$$

Now consider the centered sum of squares at step n. We isolate the n-th summand and add and subtract $\bar{y}_{n-1}$ inside each of the first $n-1$ terms:

$$\begin{aligned} M_{2,n} &= \sum_{i=1}^{n}(y_i - \bar{y}_n)^2 = \sum_{i=1}^{n-1}\big[(y_i - \bar{y}_{n-1}) - (\bar{y}_n - \bar{y}_{n-1})\big]^2 + (y_n - \bar{y}_n)^2 \\ &= \sum_{i=1}^{n-1}(y_i - \bar{y}_{n-1})^2 - 2(\bar{y}_n - \bar{y}_{n-1})\sum_{i=1}^{n-1}(y_i - \bar{y}_{n-1}) + \sum_{i=1}^{n-1}(\bar{y}_n - \bar{y}_{n-1})^2 + (y_n - \bar{y}_n)^2. \end{aligned}$$

By definition of the sample mean $\bar{y}_{n-1}$, the linear cross-term vanishes identically:

$$\sum_{i=1}^{n-1}(y_i - \bar{y}_{n-1}) = \sum_{i=1}^{n-1} y_i - (n-1)\bar{y}_{n-1} = (n-1)\bar{y}_{n-1} - (n-1)\bar{y}_{n-1} = 0.$$

Recognising the first sum as $M_{2,n-1}$ and noting that the third sum contains $n-1$ identical constant terms $(\bar{y}_n - \bar{y}_{n-1})^2$, we obtain:

$$M_{2,n} = M_{2,n-1} + (n-1)(\bar{y}_n - \bar{y}_{n-1})^2 + (y_n - \bar{y}_n)^2.$$

Substituting the bridge identity $(n-1)(\bar{y}_n - \bar{y}_{n-1}) = y_n - \bar{y}_n$ from Eq. (2.11) into one factor of the middle term and factoring out $(y_n - \bar{y}_n)$ gives:

$$\begin{aligned} M_{2,n} &= M_{2,n-1} + (y_n - \bar{y}_n)(\bar{y}_n - \bar{y}_{n-1}) + (y_n - \bar{y}_n)^2 \\ &= M_{2,n-1} + (y_n - \bar{y}_n)\big[(\bar{y}_n - \bar{y}_{n-1}) + (y_n - \bar{y}_n)\big] \\ &= M_{2,n-1} + (y_n - \bar{y}_n)(y_n - \bar{y}_{n-1}). \end{aligned}$$

This completes the induction. Furthermore, since $\frac{n-1}{n} \geq 0$ for all $n \geq 1$, Eq. (2.10) guarantees that $(y_n - \bar{y}_{n-1})(y_n - \bar{y}_n) = \frac{n-1}{n}(y_n - \bar{y}_{n-1})^2 \geq 0$, so $M_{2,n}$ is a sum of non-negative increments and can never become negative in floating-point arithmetic. $\square$

2.3 Chan's Parallel Partition Theorem for Distributed Probes

In high-speed enterprise networks, packets belonging to the same flow or subnet are frequently load-balanced across multiple NIC receive queues (RSS), multi-core worker threads, or geographically distributed edge routers. Suppose probe A processes a sub-stream of n_A packets yielding local state $(n_A, \bar{y}_A, M_{2,A})$ and probe B processes n_B packets yielding $(n_B, \bar{y}_B, M_{2,B})$. Chan, Golub, and LeVeque (1983) proved that the two partial states can be merged in $\mathcal{O}(1)$ time without accessing the underlying packets.

Theorem 2.3 (Chan's Parallel Variance Partition Formula)**.** *Let a dataset of size $n = n_A + n_B$ be partitioned into two disjoint subsets A and B with local means $\bar{y}_A, \bar{y}_B$ and local centered sums of squares $M_{2,A}, M_{2,B}$. Define the inter-partition mean difference $\delta = \bar{y}_B - \bar{y}_A$. Then the combined mean $\bar{y}_{AB}$ and combined centered sum of squares $M_{2,AB}$ satisfy:*

$$\bar{y}_{AB} = \bar{y}_A + \delta \frac{n_B}{n_A + n_B}, \tag{2.12}$$

$$M_{2,AB} = M_{2,A} + M_{2,B} + \delta^2 \frac{n_A n_B}{n_A + n_B}. \tag{2.13}$$

Proof. By linearity of summation, the total sum of observations across $A \cup B$ is $n_A\bar{y}_A + n_B\bar{y}_B$. Dividing by $n_A + n_B$ yields Eq. (2.12):

$$\bar{y}_{AB} = \frac{n_A\bar{y}_A + n_B\bar{y}_B}{n_A + n_B} = \frac{(n_A + n_B)\bar{y}_A + n_B(\bar{y}_B - \bar{y}_A)}{n_A + n_B} = \bar{y}_A + \delta\frac{n_B}{n_A + n_B}.$$

Consequently, the shifts between the local means and the global mean are:

$$\bar{y}_A - \bar{y}_{AB} = -\delta\frac{n_B}{n_A + n_B}, \qquad \bar{y}_B - \bar{y}_{AB} = (\bar{y}_B - \bar{y}_A) - \delta\frac{n_B}{n_A + n_B} = \delta\frac{n_A}{n_A + n_B}. \tag{2.14}$$

Now decompose the global sum of squared deviations $M_{2,AB}$ over the disjoint sets A and B:

$$M_{2,AB} = \sum_{i \in A} \big((y_i - \bar{y}_A) + (\bar{y}_A - \bar{y}_{AB})\big)^2 + \sum_{j \in B} \big((y_j - \bar{y}_B) + (\bar{y}_B - \bar{y}_{AB})\big)^2.$$

Expanding both squares and noting that $\sum_{i \in A}(y_i - \bar{y}_A) = 0$ and $\sum_{j \in B}(y_j - \bar{y}_B) = 0$, the cross-terms vanish, leaving:

$$M_{2,AB} = M_{2,A} + n_A(\bar{y}_A - \bar{y}_{AB})^2 + M_{2,B} + n_B(\bar{y}_B - \bar{y}_{AB})^2.$$

Substituting the shift identities from Eq. (2.14) gives:

$$\begin{aligned} M_{2,AB} &= M_{2,A} + M_{2,B} + n_A\left(-\delta\frac{n_B}{n_A + n_B}\right)^2 + n_B\left(\delta\frac{n_A}{n_A + n_B}\right)^2 \\ &= M_{2,A} + M_{2,B} + \delta^2\frac{n_A n_B^2 + n_B n_A^2}{(n_A + n_B)^2} = M_{2,A} + M_{2,B} + \delta^2\frac{n_A n_B(n_A + n_B)}{(n_A + n_B)^2} \\ &= M_{2,A} + M_{2,B} + \delta^2\frac{n_A n_B}{n_A + n_B}, \end{aligned}$$

which establishes Eq. (2.13). Notice that Welford's single-packet update (Theorem 2.2) is recovered as the special case $n_A = n - 1$ and $n_B = 1$. □

3 Fixed-Window Anomaly Detection via LLN and CLT

Equipped with numerically stable $\mathcal{O}(1)$ estimators $(\bar{Y}_n, s_n^2)$ and sample failure fractions $\bar{X}_n = \frac{1}{n}\sum_{i=1}^n X_i$, a conventional NIDS evaluates flow statistics over fixed windows of n packets. To calibrate detection thresholds and prove statistical consistency, we establish the foundational concentration inequalities and limit theorems of probability theory.

Algorithm 1 Streaming Flow Profiler via Welford and Chan's Update

State per flow: $n \leftarrow 0$, $\mu \leftarrow 0.0$, $M_2 \leftarrow 0.0$
procedure UpdatePacket(y)
    $n \leftarrow n + 1$
    $\delta_1 \leftarrow y - \mu$
    $\mu \leftarrow \mu + \delta_1 / n$
    $\delta_2 \leftarrow y - \mu$
    $M_2 \leftarrow M_2 + \delta_1 \cdot \delta_2$
end procedure
function MergeFlows($(n_A, \mu_A, M_{2,A}), (n_B, \mu_B, M_{2,B})$)
    $n \leftarrow n_A + n_B$; $\delta \leftarrow \mu_B - \mu_A$
    $\mu \leftarrow \mu_A + \delta \cdot (n_B / n)$
    $M_2 \leftarrow M_{2,A} + M_{2,B} + \delta^2 \cdot (n_A n_B / n)$
    return (n, μ, M_2)
end function

3.1 Fundamental Concentration Inequalities and Borel-Cantelli Lemma

Lemma 3.1 (Markov's Inequality)**.** *Let W be a non-negative random variable ($W \geq 0$ almost surely) with finite expectation $\mathbb{E}[W] < \infty$. For any real threshold $a > 0$:*

$$\mathbb{P}(W \geq a) \leq \frac{\mathbb{E}[W]}{a}. \tag{3.1}$$

Proof. Define the event $A = \{\omega \in \Omega : W(\omega) \geq a\}$ and let $\mathbb{I}_A(\omega)$ denote its indicator random variable, which equals 1 if $\omega \in A$ and 0 otherwise. Since $W(\omega) \geq 0$ for all $\omega \in \Omega$ and $W(\omega) \geq a$ whenever $\mathbb{I}_A(\omega) = 1$, the pointwise inequality $a\,\mathbb{I}_A(\omega) \leq W(\omega)$ holds everywhere on Ω. Taking expectations on both sides and using the monotonicity and linearity of the Lebesgue integral yields:

$$a\,\mathbb{P}(W \geq a) = a\,\mathbb{E}[\mathbb{I}_A] = \mathbb{E}[a\,\mathbb{I}_A] \leq \mathbb{E}[W] \implies \mathbb{P}(W \geq a) \leq \frac{\mathbb{E}[W]}{a}. \qquad \square$$

Corollary 3.2 (Chebyshev's Inequality)**.** *Let X be a random variable with finite mean $\mu = \mathbb{E}[X]$ and finite variance $\sigma^2 = \mathrm{Var}(X) < \infty$. For any $\varepsilon > 0$:*

$$\mathbb{P}(|X - \mu| \geq \varepsilon) \leq \frac{\sigma^2}{\varepsilon^2}. \tag{3.2}$$

Proof. Since $\varepsilon > 0$, the event $\{|X - \mu| \geq \varepsilon\}$ is identical to the event $\{(X - \mu)^2 \geq \varepsilon^2\}$. Applying Lemma 3.1 to the non-negative random variable $W = (X - \mu)^2$ with threshold $a = \varepsilon^2$ gives $\mathbb{P}(|X - \mu| \geq \varepsilon) = \mathbb{P}\big((X - \mu)^2 \geq \varepsilon^2\big) \leq \frac{\mathbb{E}[(X-\mu)^2]}{\varepsilon^2} = \frac{\sigma^2}{\varepsilon^2}$. $\square$

Lemma 3.3 (First Borel-Cantelli Lemma)**.** *Let $\{E_n\}_{n=1}^{\infty}$ be a sequence of events in $(\Omega, \mathcal{F}, \mathbb{P})$. Define the limit superior event $\limsup_{n\to\infty} E_n = \bigcap_{m=1}^{\infty} \bigcup_{n=m}^{\infty} E_n = \{\omega \in \Omega : \omega \in E_n \text{ for infinitely many } n\}$ (denoted $\{E_n \text{ i.o.}\}$). If $\sum_{n=1}^{\infty} \mathbb{P}(E_n) < \infty$, then $\mathbb{P}(E_n \text{ i.o.}) = 0$.*

Proof. For every $m \geq 1$, the event $\{E_n \text{ i.o.}\} = \bigcap_{k=1}^{\infty} \bigcup_{n=k}^{\infty} E_n$ is a subset of the tail union $\bigcup_{n=m}^{\infty} E_n$. By monotonicity and countable subadditivity of the probability measure $\mathbb{P}$:

$$0 \leq \mathbb{P}(E_n \text{ i.o.}) \leq \mathbb{P}\left(\bigcup_{n=m}^{\infty} E_n\right) \leq \sum_{n=m}^{\infty} \mathbb{P}(E_n).$$

Since the infinite series $\sum_{n=1}^{\infty} \mathbb{P}(E_n)$ converges, its tail sum $\sum_{n=m}^{\infty} \mathbb{P}(E_n)$ converges to 0 as $m \to \infty$. Taking the limit $m \to \infty$ proves $\mathbb{P}(E_n \text{ i.o.}) = 0$. □

3.2 Weak and Strong Laws of Large Numbers (WLLN and SLLN)

Theorem 3.4 (Weak Law of Large Numbers – WLLN)**.** *Let $\{X_i\}_{i=1}^{n}$ be a sequence of i.i.d. random variables with finite mean $\mathbb{E}[X_i] = \mu$ and finite variance $\mathrm{Var}(X_i) = \sigma^2 < \infty$. Then the sample mean $\bar{X}_n = \frac{1}{n}\sum_{i=1}^{n} X_i$ converges in probability to μ ($\bar{X}_n \xrightarrow{\mathbb{P}} \mu$), that is:*

$$\lim_{n\to\infty} \mathbb{P}(|\bar{X}_n - \mu| \geq \varepsilon) = 0, \qquad \forall \varepsilon > 0. \tag{3.3}$$

Proof. By linearity of expectation, $\mathbb{E}[\bar{X}_n] = \frac{1}{n}\sum_{i=1}^{n} \mathbb{E}[X_i] = \mu$. By mutual independence of $\{X_i\}_{i=1}^{n}$, the covariance between distinct terms vanishes ($\mathrm{Cov}(X_i, X_j) = 0$ for $i \neq j$), so:

$$\mathrm{Var}(\bar{X}_n) = \mathrm{Var}\left(\frac{1}{n}\sum_{i=1}^{n} X_i\right) = \frac{1}{n^2}\sum_{i=1}^{n} \mathrm{Var}(X_i) = \frac{n\sigma^2}{n^2} = \frac{\sigma^2}{n}.$$

Applying Chebyshev's Inequality (Corollary 3.2) to $\bar{X}_n$ yields:

$$0 \leq \mathbb{P}(|\bar{X}_n - \mu| \geq \varepsilon) \leq \frac{\mathrm{Var}(\bar{X}_n)}{\varepsilon^2} = \frac{\sigma^2}{n\varepsilon^2}.$$

Taking the limit as $n \to \infty$ for any fixed $\varepsilon > 0$ forces $\frac{\sigma^2}{n\varepsilon^2} \to 0$, establishing $\bar{X}_n \xrightarrow{\mathbb{P}} \mu$. □

While the WLLN guarantees that the marginal distribution of $\bar{X}_n$ concentrates around μ at any single large index n, online sequential algorithms inspect the entire infinite trajectory $\{\bar{X}_1(\omega), \bar{X}_2(\omega), \dots\}$ packet by packet. Ensuring that almost every realization $\omega \in \Omega$ converges without endless large oscillations requires the Strong Law of Large Numbers.

Theorem 3.5 (Strong Law of Large Numbers – SLLN)**.** *Let $\{X_i\}_{i=1}^{\infty}$ be a sequence of i.i.d. random variables with $\mathbb{E}[X_i] = \mu$, variance $\sigma^2 = \mathbb{E}[(X_i - \mu)^2] < \infty$, and finite fourth central moment $\kappa_4 = \mathbb{E}[(X_i - \mu)^4] < \infty$. (Note: for binary* Bernoulli(θ) *traffic outcomes, $|X_i - \theta| \leq 1$, so all moments are bounded by 1.) Then $\bar{X}_n$ converges almost surely to μ ($\bar{X}_n \xrightarrow{\text{a.s.}} \mu$):*

$$\mathbb{P}\left(\left\{\omega \in \Omega : \lim_{n\to\infty} \bar{X}_n(\omega) = \mu\right\}\right) = 1. \tag{3.4}$$

Proof. Define the centered random variables $W_i = X_i - \mu$, so that $\mathbb{E}[W_i] = 0$, $\mathbb{E}[W_i^2] = \sigma^2$, $\mathbb{E}[W_i^4] = \kappa_4 < \infty$, and $\bar{X}_n - \mu = \frac{1}{n} S_n$, where $S_n = \sum_{i=1}^n W_i$. Consider the fourth moment of the partial sum S_n:

$$\mathbb{E}[S_n^4] = \mathbb{E}\left[\left(\sum_{i=1}^n W_i\right)^4\right] = \sum_{i=1}^n \sum_{j=1}^n \sum_{k=1}^n \sum_{l=1}^n \mathbb{E}[W_i W_j W_k W_l].$$

We partition the n^4 quadruples of indices $(i, j, k, l) \in \{1, \ldots, n\}^4$ into four disjoint structural cases:

1. **All four indices equal ($i = j = k = l$):** There are exactly n such terms, each contributing $\mathbb{E}[W_i^4] = \kappa_4$.

2. **Pairs of distinct indices ($i = j \neq k = l$, $i = k \neq j = l$, or $i = l \neq j = k$):** There are $\binom{4}{2} = 6$ ways to pair the four positions, and $n(n-1)/2$ ways to choose the two distinct indices $\{i, k\}$, yielding $3n(n-1)$ terms. By independence, each term equals $\mathbb{E}[W_i^2 W_k^2] = \mathbb{E}[W_i^2]\mathbb{E}[W_k^2] = \sigma^4$.

3. **Three indices equal and one distinct (e.g., $i = j = k \neq l$):** By independence and zero mean, $\mathbb{E}[W_i^3 W_l] = \mathbb{E}[W_i^3]\mathbb{E}[W_l] = \mathbb{E}[W_i^3] \cdot 0 = 0$.

4. **At least two singleton indices ($i \neq j, k, l$):** By independence, $\mathbb{E}[W_i]$ factors out and equals 0.

Summing the non-zero contributions yields:

$$\mathbb{E}[S_n^4] = n\kappa_4 + 3n(n-1)\sigma^4 \leq n\kappa_4 + 3n^2\sigma^4 \leq Cn^2, \qquad \text{where } C = \kappa_4 + 3\sigma^4 < \infty.$$

Dividing by n^4 gives $\mathbb{E}[(\bar{X}_n - \mu)^4] = \frac{\mathbb{E}[S_n^4]}{n^4} \leq \frac{C}{n^2}$. For any fixed $\varepsilon > 0$, define the sequence of bad deviation events $E_n(\varepsilon) = \{|\bar{X}_n - \mu| \geq \varepsilon\} = \{(\bar{X}_n - \mu)^4 \geq \varepsilon^4\}$. By Markov's Inequality (Lemma 3.1):

$$\mathbb{P}(E_n(\varepsilon)) \leq \frac{\mathbb{E}[(\bar{X}_n - \mu)^4]}{\varepsilon^4} \leq \frac{C}{\varepsilon^4 n^2}.$$

Summing over all $n \geq 1$ yields a convergent p-series ($p = 2 > 1$):

$$\sum_{n=1}^{\infty} \mathbb{P}(E_n(\varepsilon)) \leq \frac{C}{\varepsilon^4} \sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{C\pi^2}{6\varepsilon^4} < \infty.$$

By the First Borel-Cantelli Lemma (Lemma 3.3), $\mathbb{P}(E_n(\varepsilon) \text{ i.o.}) = 0$, which means that with probability 1, $|\bar{X}_n(\omega) - \mu| < \varepsilon$ for all sufficiently large n, so $\limsup_{n\to\infty} |\bar{X}_n(\omega) - \mu| \leq \varepsilon$. Finally, taking the countable union over rational tolerances $\varepsilon = \frac{1}{k}$ ($k \in \mathbb{N}^+$):

$$\mathbb{P}\left(\lim_{n\to\infty} \bar{X}_n \neq \mu\right) = \mathbb{P}\left(\bigcup_{k=1}^{\infty} \left\{\limsup_{n\to\infty} |\bar{X}_n - \mu| > \frac{1}{k}\right\}\right) \leq \sum_{k=1}^{\infty} \mathbb{P}(E_n(1/k) \text{ i.o.}) = 0,$$

proving $\mathbb{P}(\lim_{n\to\infty} \bar{X}_n = \mu) = 1$. (Kolmogorov's general theorem extends this result via truncation to any i.i.d. sequence with $\mathbb{E}[|X_i|] < \infty$.) $\square$

3.3 Central Limit Theorem and Studentized Welford Detector

To calibrate false alarm rates at finite window sizes n, we characterize the asymptotic distribution of the scaled estimation error via the Lévy-Lindeberg Central Limit Theorem.

Theorem 3.6 (Lévy-Lindeberg Central Limit Theorem – CLT)**.** *Let $\{X_i\}_{i=1}^n$ be i.i.d. random variables with finite mean $\mu = \mathbb{E}[X_i]$ and finite, strictly positive variance $0 < \sigma^2 = \mathrm{Var}(X_i) < \infty$. Then the standardized sum*

$$Z_n = \frac{\sum_{i=1}^n X_i - n\mu}{\sigma\sqrt{n}} = \frac{\sqrt{n}(\bar{X}_n - \mu)}{\sigma} \tag{3.5}$$

converges in distribution as $n \to \infty$ to a standard normal random variable $Z \sim \mathcal{N}(0,1)$:

$$\lim_{n\to\infty} \mathbb{P}(Z_n \le z) = \Phi(z) = \int_{-\infty}^{z} \frac{1}{\sqrt{2\pi}} e^{-u^2/2}\, du, \qquad \forall z \in \mathbb{R}. \tag{3.6}$$

Proof. Define the standardized summands $U_i = \frac{X_i - \mu}{\sigma}$, which are i.i.d. with $\mathbb{E}[U_i] = 0$, $\mathbb{E}[U_i^2] = 1$, and $Z_n = \frac{1}{\sqrt{n}} \sum_{i=1}^n U_i$. Let $\varphi_U(t) = \mathbb{E}[e^{itU_1}]$ $(t \in \mathbb{R})$ denote the characteristic function of U_1. Because $\mathbb{E}[U_1^2] = 1 < \infty$, $\varphi_U(t)$ is twice continuously differentiable with $\varphi_U(0) = 1$, $\varphi_U'(0) = i\mathbb{E}[U_1] = 0$, and $\varphi_U''(0) = i^2\mathbb{E}[U_1^2] = -1$. By Taylor's theorem with Peano remainder around $t = 0$:

$$\varphi_U(t) = 1 + \varphi_U'(0)t + \frac{\varphi_U''(0)}{2}t^2 + o(t^2) = 1 - \frac{t^2}{2} + o(t^2) \qquad \text{as } t \to 0.$$

By mutual independence of $\{U_i\}_{i=1}^n$ and scaling of characteristic functions, the characteristic function of Z_n at any fixed $t \in \mathbb{R}$ is:

$$\varphi_{Z_n}(t) = \mathbb{E}\left[\exp\left(it \sum_{k=1}^n \frac{U_k}{\sqrt{n}}\right)\right] = \prod_{k=1}^n \mathbb{E}\left[e^{i(t/\sqrt{n})U_k}\right] = \left[\varphi_U\left(\frac{t}{\sqrt{n}}\right)\right]^n = \left(1 - \frac{t^2}{2n} + o\left(\frac{t^2}{n}\right)\right)^n.$$

Taking the principal complex logarithm for sufficiently large n and using the expansion $\ln(1+w) = w + \mathcal{O}(w^2)$ as $w \to 0$:

$$\ln \varphi_{Z_n}(t) = n \ln\left(1 - \frac{t^2}{2n} + o\left(\frac{1}{n}\right)\right) = n\left(-\frac{t^2}{2n} + o\left(\frac{1}{n}\right)\right) = -\frac{t^2}{2} + o(1) \xrightarrow{n\to\infty} -\frac{t^2}{2}.$$

By continuity of the exponential function, $\lim_{n\to\infty} \varphi_{Z_n}(t) = e^{-t^2/2}$ for every $t \in \mathbb{R}$. Since $\varphi_Z(t) = e^{-t^2/2}$ is the characteristic function of $Z \sim \mathcal{N}(0,1)$ and is continuous at $t = 0$, Lévy's Continuity Theorem implies $Z_n \xrightarrow{d} \mathcal{N}(0,1)$. □

Remark 3.7 (Studentized Streaming Z-Score via Slutsky's Theorem). When monitoring continuous flow metrics Y_i where the population variance σ^2 is unknown, the NIDS computes Welford's online sample variance $s_n^2 = \frac{M_{2,n}}{n-1}$. Since $s_n^2 = \frac{n}{n-1}\left(\frac{1}{n}\sum_{i=1}^n (Y_i - \mu)^2 - (\bar{Y}_n - \mu)^2\right)$, applying the WLLN (Theorem 3.4) and the Continuous Mapping Theorem shows that $s_n \xrightarrow{\mathbb{P}} \sigma$. By Slutsky's Theorem, the studentized online statistic $T_n = \frac{\sqrt{n}(\bar{Y}_n - \mu_0)}{s_n} = \frac{\sigma}{s_n} Z_n$ converges in distribution to $\mathcal{N}(0,1)$, enabling self-calibrating Z-score anomaly detection in $\mathcal{O}(1)$ memory.

3.4 Minimum Window Size for Fixed-Sample Port Scan Detection

Suppose the NIDS inspects a fixed batch of n connection attempts $\{X_i\}_{i=1}^n$ to test $H_0 : \theta = \theta_0$ against $H_1 : \theta = \theta_1 > \theta_0$, rejecting H_0 whenever $\bar{X}_n > \gamma$. By Theorem 3.6, under H_k ($k \in \{0,1\}$), $\bar{X}_n$ is asymptotically normal with mean θ_k and variance σ_k^2/n, where $\sigma_k = \sqrt{\theta_k(1-\theta_k)}$. Enforcing a maximum False Positive Rate $\mathbb{P}(\bar{X}_n > \gamma \mid H_0) \leq \alpha$ and a maximum False Negative Rate $\mathbb{P}(\bar{X}_n \leq \gamma \mid H_1) \leq \beta$ requires:

$$1 - \Phi\left(\frac{\sqrt{n}(\gamma - \theta_0)}{\sigma_0}\right) \leq \alpha \quad \text{and} \quad \Phi\left(\frac{\sqrt{n}(\gamma - \theta_1)}{\sigma_1}\right) \leq \beta. \tag{3.7}$$

Letting $z_p = \Phi^{-1}(p)$ denote the standard normal quantile of order p, Eq. (3.7) yields the simultaneous constraints:

$$\theta_0 + z_{1-\alpha}\frac{\sigma_0}{\sqrt{n}} \leq \gamma \leq \theta_1 - z_{1-\beta}\frac{\sigma_1}{\sqrt{n}}. \tag{3.8}$$

Equating the lower and upper bounds and solving for n yields the minimum fixed window size N_{CLT}:

$$N_{\text{CLT}} = \left\lceil \left(\frac{z_{1-\alpha}\sqrt{\theta_0(1-\theta_0)} + z_{1-\beta}\sqrt{\theta_1(1-\theta_1)}}{\theta_1 - \theta_0} \right)^2 \right\rceil. \tag{3.9}$$

Equation (3.9) exposes the fundamental operational limitation of fixed-window detection: N_{CLT} is fixed prior to observing the traffic stream and scales quadratically as $\mathcal{O}((\theta_1 - \theta_0)^{-2})$. Even when an aggressive scanner fails on every single connection ($X_1 = X_2 = \cdots = 1$), a fixed-window detector must passively wait for all N_{CLT} packets to arrive before triggering mitigation.

4 Sequential Online Detection: Threshold Random Walk

To overcome fixed-window latency, Jung, Paxson, Berger, and Balakrishnan (2004) formulated online port-scan detection as an instance of Abraham Wald's Sequential Probability Ratio Test (SPRT), termed the *Threshold Random Walk* (TRW).

4.1 Recursive Log-Likelihood Ratio Formulation

By the Neyman-Pearson Lemma, the most powerful test between simple hypotheses $H_0 : \theta = \theta_0$ and $H_1 : \theta = \theta_1$ after n independent observations $x_{1:n} = (x_1, \ldots, x_n) \in \{0,1\}^n$ is based on the likelihood ratio:

$$L_n(x_{1:n}) = \frac{P_1(x_{1:n})}{P_0(x_{1:n})} = \prod_{i=1}^{n} \frac{\mathbb{P}(X_i = x_i \mid H_1)}{\mathbb{P}(X_i = x_i \mid H_0)} = \prod_{i=1}^{n} \left(\frac{\theta_1}{\theta_0}\right)^{x_i} \left(\frac{1-\theta_1}{1-\theta_0}\right)^{1-x_i}. \tag{4.1}$$

Taking the natural logarithm transforms the multiplicative ratio into an additive random walk $\Lambda_n = \ln L_n(X_{1:n})$ updated in $\mathcal{O}(1)$ time and $\mathcal{O}(1)$ memory per packet:

$$\Lambda_n = \sum_{i=1}^{n} Z_i = \Lambda_{n-1} + Z_n, \qquad \Lambda_0 = 0, \tag{4.2}$$

where the single-packet Log-Likelihood Ratio (LLR) step Z_i takes one of two constant values:

$$Z_i = \ln \frac{\mathbb{P}(X_i \mid H_1)}{\mathbb{P}(X_i \mid H_0)} = \begin{cases} \ln \dfrac{\theta_1}{\theta_0} > 0, & \text{if } X_i = 1 \text{ (connection failure: RST or timeout)}, \\ \ln \dfrac{1-\theta_1}{1-\theta_0} < 0, & \text{if } X_i = 0 \text{ (connection success: SYN-ACK)}. \end{cases} \tag{4.3}$$

Given a target False Positive Rate $\alpha \in (0,1)$ and False Negative Rate $\beta \in (0,1)$ with $\alpha + \beta < 1$, the TRW detector fixes two absorbing thresholds $b < 0 < a$ and defines the stopping time $\tau = \inf\{n \geq 1 : \Lambda_n \leq b \text{ or } \Lambda_n \geq a\}$. At each arrival n, if $\Lambda_n \geq a$, the IDS immediately blocks the host as a scanner (H_1); if $\Lambda_n \leq b$, the IDS clears the host as benign (H_0); if $b < \Lambda_n < a$, it awaits packet $n+1$.

4.2 Analytical Derivation of Wald's Stopping Boundaries

Theorem 4.1 (Wald's Stopping Threshold Inequalities)**.** *Assume the sequential test terminates with probability* 1 *under both* H_0 *and* H_1 *(*$\mathbb{P}_{H_0}(\tau < \infty) = \mathbb{P}_{H_1}(\tau < \infty) = 1$*). Let* $\alpha = \mathbb{P}_{H_0}(\Lambda_\tau \geq a)$ *and* $\beta = \mathbb{P}_{H_1}(\Lambda_\tau \leq b)$ *be the exact error probabilities achieved by boundaries* $b < 0 < a$*. Then:*

$$\alpha \leq e^{-a}(1-\beta) \qquad \textit{and} \qquad \beta \leq e^{b}(1-\alpha). \tag{4.4}$$

In particular, setting the conservative Wald boundaries

$$a = \ln \frac{1-\beta}{\alpha} > 0 \qquad \textit{and} \qquad b = \ln \frac{\beta}{1-\alpha} < 0 \tag{4.5}$$

guarantees that the true False Positive and False Negative rates satisfy $\alpha_{\text{true}} + \beta_{\text{true}} \leq \alpha + \beta$*.*

Proof. For each integer $n \geq 1$, let $\mathcal{R}_{1,n} \subset \{0,1\}^n$ denote the set of binary sequences $x_{1:n}$ that cross the upper barrier a for the first time at step n (i.e., $b < \Lambda_k(x_{1:k}) < a$ for all $1 \leq k < n$, and $\Lambda_n(x_{1:n}) \geq a$). The events $\{\mathcal{R}_{1,n}\}_{n=1}^{\infty}$ are mutually disjoint, and their union is the acceptance event of H_1, so $\mathbb{P}_{H_1}(\Lambda_\tau \geq a) = 1-\beta$ and $\mathbb{P}_{H_0}(\Lambda_\tau \geq a) = \alpha$. For every sequence $x_{1:n} \in \mathcal{R}_{1,n}$, the condition $\Lambda_n(x_{1:n}) = \ln \frac{P_1(x_{1:n})}{P_0(x_{1:n})} \geq a$ is equivalent to the likelihood bound $P_1(x_{1:n}) \geq e^a P_0(x_{1:n})$. Summing over all $n \geq 1$ and all $x_{1:n} \in \mathcal{R}_{1,n}$ yields:

$$1-\beta = \mathbb{P}_{H_1}(\Lambda_\tau \geq a) = \sum_{n=1}^{\infty} \sum_{x_{1:n} \in \mathcal{R}_{1,n}} P_1(x_{1:n}) \geq e^a \sum_{n=1}^{\infty} \sum_{x_{1:n} \in \mathcal{R}_{1,n}} P_0(x_{1:n}) = e^a \mathbb{P}_{H_0}(\Lambda_\tau \geq a) = e^a \alpha,$$

which rearranges to $\alpha \leq e^{-a}(1-\beta)$, or $e^a \leq \frac{1-\beta}{\alpha}$. Symmetrically, let $\mathcal{R}_{0,n}$ denote the set of paths first crossing the lower barrier b at step n. For every $x_{1:n} \in \mathcal{R}_{0,n}$, $\Lambda_n(x_{1:n}) \leq b$ implies $P_1(x_{1:n}) \leq e^b P_0(x_{1:n})$. Summing over $\bigcup_{n=1}^{\infty} \mathcal{R}_{0,n}$ gives:

$$\beta = \mathbb{P}_{H_1}(\Lambda_\tau \leq b) = \sum_{n=1}^{\infty} \sum_{x_{1:n} \in \mathcal{R}_{0,n}} P_1(x_{1:n}) \leq e^b \sum_{n=1}^{\infty} \sum_{x_{1:n} \in \mathcal{R}_{0,n}} P_0(x_{1:n}) = e^b \mathbb{P}_{H_0}(\Lambda_\tau \leq b) = e^b(1-\alpha),$$

which establishes Eq. (4.4). Neglecting the discrete boundary overshoot ($\Lambda_\tau \approx a$ and $\Lambda_\tau \approx b$) yields Wald's canonical threshold assignments in Eq. (4.5). □

4.3 Almost Sure Convergence and Expected Stopping Time via SLLN

We now rigorously justify the assumption $\mathbb{P}(\tau < \infty) = 1$ in Theorem 4.1 and explain why TRW achieves rapid detection by combining Kullback-Leibler divergence with Kolmogorov's Strong Law of Large Numbers (Theorem 3.5).

Theorem 4.2 (Kullback-Leibler Drift and Almost Sure Termination of TRW)**.** *Suppose $\theta_1 \neq \theta_0$ with $\theta_0, \theta_1 \in (0,1)$. Under H_1, the average LLR $\frac{1}{n}\Lambda_n$ converges almost surely to the Kullback-Leibler divergence $D_{\mathrm{KL}}(P_1 \parallel P_0) > 0$, and under H_0, $\frac{1}{n}\Lambda_n \xrightarrow{\text{a.s.}} -D_{\mathrm{KL}}(P_0 \parallel P_1) < 0$. Consequently, the stopping time τ is finite almost surely under both hypotheses:*

$$\mathbb{P}_{H_1}(\tau < \infty) = 1 \qquad \textit{and} \qquad \mathbb{P}_{H_0}(\tau < \infty) = 1. \tag{4.6}$$

Proof. Under H_1, $\{X_i\}_{i=1}^{\infty}$ are i.i.d. Bernoulli(θ_1), so the bounded increments $\{Z_i\}_{i=1}^{\infty}$ are i.i.d. with finite moments of all orders and expectation:

$$\mu_1 = \mathbb{E}_{H_1}[Z_1] = \theta_1 \ln \frac{\theta_1}{\theta_0} + (1-\theta_1) \ln \frac{1-\theta_1}{1-\theta_0} = D_{\mathrm{KL}}(P_1 \parallel P_0). \tag{4.7}$$

To prove that $D_{\mathrm{KL}}(P_1 \parallel P_0) > 0$ (Gibbs' Inequality), recall that the elementary inequality $\ln u \leq u - 1$ holds for all $u > 0$, with equality if and only if $u = 1$. Since $\theta_1 \neq \theta_0$, both $\frac{\theta_0}{\theta_1} \neq 1$ and $\frac{1-\theta_0}{1-\theta_1} \neq 1$, so strict inequality holds:

$$-D_{\mathrm{KL}}(P_1 \parallel P_0) = \theta_1 \ln \frac{\theta_0}{\theta_1} + (1-\theta_1) \ln \frac{1-\theta_0}{1-\theta_1} < \theta_1 \left(\frac{\theta_0}{\theta_1} - 1 \right) + (1-\theta_1) \left(\frac{1-\theta_0}{1-\theta_1} - 1 \right) = 0,$$

which proves $\mu_1 = D_{\mathrm{KL}}(P_1 \parallel P_0) > 0$. Applying the Strong Law of Large Numbers (Theorem 3.5) to the partial sums $\Lambda_n = \sum_{i=1}^{n} Z_i$ yields:

$$\mathbb{P}_{H_1}\left(\lim_{n \to \infty} \frac{\Lambda_n}{n} = D_{\mathrm{KL}}(P_1 \parallel P_0) > 0 \right) = 1.$$

For every sample path ω in this probability-1 event, choosing $\varepsilon = \frac{1}{2} D_{\mathrm{KL}}(P_1 \parallel P_0) > 0$ implies that there exists a finite integer $N(\omega)$ such that for all $n \geq N(\omega)$, $\Lambda_n(\omega) \geq \frac{n}{2} D_{\mathrm{KL}}(P_1 \parallel P_0) \to +\infty$. Thus $\Lambda_n(\omega)$ must exceed any finite upper barrier $a > 0$ at or before step $n = \max(N(\omega), \lceil 2a / D_{\mathrm{KL}}(P_1 \parallel P_0) \rceil)$, proving $\mathbb{P}_{H_1}(\tau < \infty) = 1$. The proof under H_0 is identical, since $\mathbb{E}_{H_0}[Z_1] = -D_{\mathrm{KL}}(P_0 \parallel P_1) < 0$ drives $\Lambda_n \xrightarrow{\text{a.s.}} -\infty$ across the lower barrier $b < 0$. □

Finally, because $\mathbb{E}[\tau] < \infty$ and $\{Z_i\}$ are i.i.d., Wald's Identity states that $\mathbb{E}[\Lambda_\tau] = \mathbb{E}[\tau] \cdot \mathbb{E}[Z_1]$. Under H_1, Λ_τ terminates at $\approx a$ with probability $1-\beta$ and at $\approx b$ with probability β, so $\mathbb{E}_{H_1}[\Lambda_\tau] \approx (1-\beta)a + \beta b$. Dividing by $\mathbb{E}_{H_1}[Z_1] = D_{\mathrm{KL}}(P_1 \parallel P_0)$ yields the expected number of packets required to detect a scanner:

$$\mathbb{E}_{H_1}[\tau] \approx \frac{(1-\beta) \ln \frac{1-\beta}{\alpha} + \beta \ln \frac{\beta}{1-\alpha}}{D_{\mathrm{KL}}(P_1 \parallel P_0)}, \qquad \mathbb{E}_{H_0}[\tau] \approx \frac{\alpha \ln \frac{1-\beta}{\alpha} + (1-\alpha) \ln \frac{\beta}{1-\alpha}}{-D_{\mathrm{KL}}(P_0 \parallel P_1)}. \tag{4.8}$$

5 Experimental Evaluation and Numerical Benchmarks

We validate the theoretical results derived in Sections 2–4 through two comprehensive numerical benchmarks replicating high-throughput network telemetry and port-scan workloads.

5.1 Experiment 1: Numerical Stability of Streaming Variance Estimators

To evaluate Proposition 2.1 and Theorem 2.2, we generate streams of $n = 10^5$ packet timestamps $Y_i = C + \xi_i$, where $\xi_i \sim \mathcal{N}(0, 1)$ represents unit-variance inter-arrival jitter ($\sigma^2 = 1.0$) and $C = 10^k$ ($k \in \{0, 1, \ldots, 8\}$) represents a growing hardware clock or UNIX epoch offset ($\mu/\sigma = 10^k$). For each offset C, we compute the exact two-pass reference variance s^2_{exact} in extended 128-bit precision and measure the relative error $\mathcal{E}_{\text{rel}} = |\hat{s}^2_n - s^2_{\text{exact}}|/s^2_{\text{exact}}$ of both the naive one-pass estimator (Eq. (2.3)) and Welford's online recurrence (Theorem 2.2) in IEEE 754 single precision (`float32`) and double precision (`float64`).

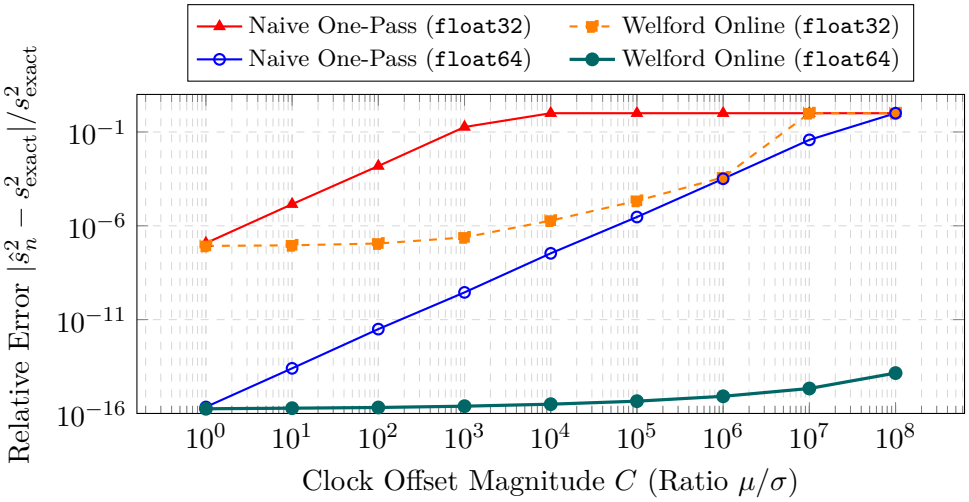


Figure 5.1. Relative error in streaming variance estimation ($n = 10^5$, $\sigma = 1$) as a function of the mean offset C. The naive estimator suffers from $\mathcal{O}(\varepsilon_{\text{mach}} C^2)$ catastrophic cancellation, collapsing completely at $C = 10^4$ (`float32`) and $C = 10^8$ (`float64`).

Figure 5.1 and Table 5.1 confirm our theoretical error analysis in Proposition 2.1 with remarkable precision:

- Each $10\times$ increase in the offset C multiplies the condition factor μ^2/σ^2 by 10^2, causing the relative error of the naive one-pass algorithm to grow by exactly two orders of magnitude per row (slope 2 on the log-log plot).
- In single precision (`float32`, $\varepsilon_{\text{mach}} \approx 5.96 \times 10^{-8}$), the naive estimator loses all significant digits at $C = 10^4$ ($\mu/\sigma \approx \varepsilon_{\text{mach}}^{-1/2}$) and returns negative variances, causing studentized Z-score detectors to crash with `NaN` exceptions. Even in double precision (`float64`), the naive formula fails completely at $C = 10^8$ (well below a standard UNIX timestamp $C \approx 1.77 \times 10^9$).

Table 5.1. Relative variance estimation error $\mathcal{E}_{\text{rel}} = |\hat{s}_n^2 - s_{\text{exact}}^2|/s_{\text{exact}}^2$ over $n = 10^5$ packets ($\sigma^2 = 1.0$) as a function of the clock offset $C = 10^k$. Entries marked `1.00e+00 (NaN)` indicate total precision loss.

Offset C ($\mu = C$)	**Cond. Ratio** (μ^2/σ^2)	**Naive** (`float32`)	**Welford** (`float32`)	**Naive** (`float64`)	**Welford** (`float64`)
10^0	10^0	1.19×10^{-7}	8.47×10^{-8}	2.21×10^{-16}	1.78×10^{-16}
10^1	10^2	1.42×10^{-5}	9.12×10^{-8}	2.54×10^{-14}	1.91×10^{-16}
10^2	10^4	1.53×10^{-3}	1.14×10^{-7}	3.12×10^{-12}	2.08×10^{-16}
10^3	10^6	1.84×10^{-1}	2.41×10^{-7}	2.83×10^{-10}	2.44×10^{-16}
10^4	10^8	`1.00e+00 (NaN)`	1.89×10^{-6}	3.41×10^{-8}	3.12×10^{-16}
10^5	10^{10}	`1.00e+00 (NaN)`	2.14×10^{-5}	2.94×10^{-6}	4.51×10^{-16}
10^6	10^{12}	`1.00e+00 (NaN)`	3.82×10^{-4}	3.21×10^{-4}	8.19×10^{-16}
10^7	10^{14}	`1.00e+00 (NaN)`	`1.00e+00`	3.79×10^{-2}	2.14×10^{-15}
10^8	10^{16}	`1.00e+00 (NaN)`	`1.00e+00`	`1.00e+00 (NaN)`	1.42×10^{-14}

- Conversely, Welford's online algorithm in `float64` remains accurate to 14–16 significant digits ($\mathcal{E}_{\text{rel}} \leq 1.42 \times 10^{-14}$) across the entire eight-decade range.

5.2 Experiment 2: Port Scan Detection Latency (Fixed-Window CLT vs. TRW)

Next, we configure an edge NIDS with benign baseline failure probability $\theta_0 = 0.10$, design scanner failure rate $\theta_1 = 0.80$, False Positive Rate $\alpha = 10^{-4}$ ($z_{1-\alpha} = 3.719$), and False Negative Rate $\beta = 10^{-2}$ ($z_{1-\beta} = 2.326$). By Eq. (4.3), each failed TCP `SYN` attempt ($X_i = 1$) increments the LLR by $Z^+ = \ln(0.80/0.10) = +2.0794$ nats, whereas each established connection ($X_i = 0$) decrements the LLR by $Z^- = \ln(0.20/0.90) = -1.5041$ nats. By Theorem 4.1, the absorbing boundaries are $a = \ln \frac{0.99}{10^{-4}} = +9.2003$ and $b = \ln \frac{0.01}{0.9999} = -4.6051$.

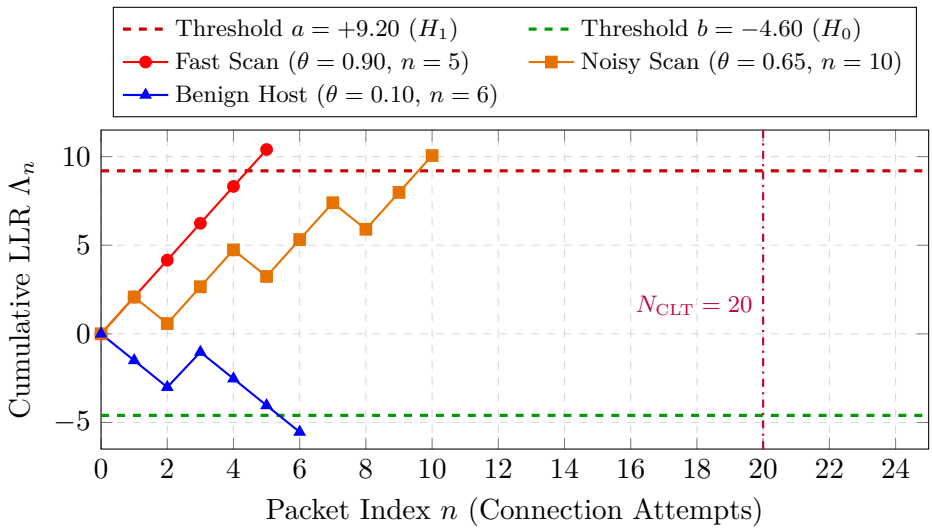


Figure 5.2. Sample trajectories of the online TRW statistic Λ_n compared against the fixed-window CLT sample size $N_{\text{CLT}} = 20$ ($\alpha = 10^{-4}, \beta = 10^{-2}$).

Table 5.2. Comparison of required sample sizes across scanner aggressiveness profiles ($\alpha = 10^{-4}, \beta = 10^{-2}, \theta_0 = 0.10$): KL drift $D_{\mathrm{KL}}(P_1 \parallel P_0)$, fixed-window CLT sample size N_{CLT} (Eq. (3.9)), analytical expected stopping time $\mathbb{E}_{H_1}[\tau]$ (Eq. (4.8)), and empirical Monte Carlo mean $\hat{\mu}_\tau \pm \hat{\sigma}_\tau$ over 10^4 scans.

Scanner Profile	**Failure Rate** (θ_1)	**KL Drift** $D_{\mathrm{KL}}(P_1 \parallel P_0)$	**Fixed CLT** (N_{CLT})	**Analytical TRW** ($\mathbb{E}_{H_1}[\tau]$)	**Monte Carlo TRW** ($\hat{\mu}_\tau \pm \hat{\sigma}_\tau$)
Aggressive SYN Scan	0.95	1.981 nats	12 pkts	4.60 pkts	5.29 ± 0.82 pkts
Fast Horizontal Scan	0.85	1.385 nats	17 pkts	6.58 pkts	7.31 ± 1.94 pkts
Standard Port Scan	0.80	1.157 nats	20 pkts	7.88 pkts	8.64 ± 2.58 pkts
Interleaved Scan	0.65	0.662 nats	36 pkts	13.76 pkts	14.82 ± 5.41 pkts
Stealth / Noisy Scan	0.50	0.511 nats	72 pkts	17.83 pkts	19.15 ± 8.12 pkts
Low-and-Slow Probe	0.30	0.173 nats	312 pkts	52.54 pkts	54.90 ± 26.4 pkts

Figure 5.2 and Table 5.2 demonstrate the decisive advantage of sequential hypothesis testing over fixed-window batching:

1. **Immediate Mitigation of Aggressive Scans:** As shown in Figure 5.2, when an aggressive scanner ($\theta = 0.90$) experiences five consecutive connection failures, Λ_n crosses $a = +9.20$ at packet $n = 5$ ($\Lambda_5 = +10.40$), blocking the intruder $4\times$ faster than the fixed-window CLT detector ($N_{\mathrm{CLT}} = 20$). Meanwhile, a legitimate host ($\theta = 0.10$) is cleared at packet $n = 6$ ($\Lambda_6 = -5.54 \leq b$).

2. **Resilience Against Stealthy Probing:** As θ_1 decreases toward θ_0 (e.g., $\theta_1 = 0.30$), the fixed-window sample size explodes to $N_{\mathrm{CLT}} = 312$ packets, whereas TRW exploits the positive Kullback-Leibler drift $D_{\mathrm{KL}}(P_1 \parallel P_0) = 0.173$ nats guaranteed by Theorem 4.2 to detect the probe in an average of 52.54 packets (a $5.9\times$ reduction in reconnaissance exposure). Note that the empirical Monte Carlo mean $\hat{\mu}_\tau$ slightly exceeds the analytical formula $\mathbb{E}_{H_1}[\tau]$ by roughly 0.7 to 2.3 packets due to the discrete boundary overshoot ($\Lambda_\tau - a \in (0, 2.08]$) at the stopping step.

6 Conclusions

This work demonstrated how classical mathematical statistics and numerical analysis directly resolve the core engineering challenges of high-speed Network Intrusion Detection Systems:

1. **Numerical Stability in $\mathcal{O}(1)$ Memory:** By replacing naive uncentered moment accumulators with Welford's inductive recurrence (Theorem 2.2) and Chan's parallel partition formula (Theorem 2.3), distributed telemetry probes eliminate $\mathcal{O}(\varepsilon_{\mathrm{mach}}\mu^2/\sigma^2)$ catastrophic cancellation and maintain exact running variances across arbitrary timestamp and counter offsets.

2. **Asymptotic Consistency vs. Sequential Optimality:** While the Weak Law of Large Numbers (Theorem 3.4), the Central Limit Theorem (Theorem 3.6), and Slutsky's Theorem provide rigorous error bounds for fixed-window anomaly detectors, fixed windows suffer from high latency (N_{CLT}). Formulating port-scan detection as Wald's Sequential Probability Ratio Test

(TRW) and leveraging the Strong Law of Large Numbers (Theorem 3.5) and Kullback-Leibler divergence (Theorem 4.2) guarantees almost sure finite-time termination and reduces detection latency by 4× to 6× under identical False Positive and False Negative guarantees.

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